

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS1

SUBJECT: Biology

TOPIC: Nutrition in Plants (Photosynthesis)

Introduction

- Nutrition is the process by which organisms obtain and use food. Plants are autotrophic: they make their own food by photosynthesis, using light energy, carbon dioxide and water.

The process of photosynthesis

- Carbon dioxide + water → glucose + oxygen (in the presence of light and chlorophyll).
- Chlorophyll in the chloroplasts traps light energy; the light reaction splits water (photolysis) releasing oxygen; the dark reactions use hydrogen and CO₂ to build glucose.
- The glucose is converted to starch for storage, and used in respiration to release energy.
- Chloroplasts are found mainly in the palisade mesophyll of the leaf; stomata admit CO₂, and the xylem brings water.

Why photosynthesis matters

- It is the only process that changes light energy into chemical energy stored in food; it is the source of oxygen in the atmosphere and the basis of every food chain.

CLASS EVALUATION

1. Write the balanced word equation for photosynthesis.
2. State the role of chlorophyll.
3. What is the importance of photosynthesis to other living things?
4. Where in the leaf does most photosynthesis occur and why?

NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS1

SUBJECT: Biology

TOPIC: Factors Affecting Photosynthesis & Plant Nutrition

Introduction

- Photosynthesis is a rate process: its speed depends on several factors, and it stops when any of them is missing (the limiting factor concept).

Limiting factors

- **Light** — more light up to a saturation point raises the rate; in darkness no photosynthesis occurs.
- **Carbon dioxide** — the normal 0.04% in air is the main limiting factor; greenhouse growers enrich CO₂.
- **Temperature** — enzyme-controlled; rate rises with warmth but falls above about 45 °C when enzymes denature.
- **Water** — shortage closes stomata and slows the supply of CO₂.

Mineral nutrition of plants

- **Macronutrients** — nitrogen (leaves and growth), phosphorus (roots, energy transfer), potassium (water balance, flowering), magnesium (chlorophyll), calcium.
- **Micronutrients** — iron, zinc, copper, boron, manganese, needed in trace amounts.
- Deficiencies show as yellowing (chlorosis), stunted growth and poor flowering; fertilisers replace lost minerals.

CLASS EVALUATION

1. Define a limiting factor with an example.
2. Why does photosynthesis stop at very high temperatures?
3. Name two macronutrients and their functions.
4. What is chlorosis and what causes it?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS1

SUBJECT: Biology

TOPIC: Nutrition in Animals (Heterotrophic Nutrition)

Introduction

- Animals cannot make their own food; they are heterotrophs — they feed on other organisms. Heterotrophic nutrition has four main types.

Types of heterotrophic nutrition

- **Holozoic** — feeding on solid organic matter which is ingested, digested, absorbed and assimilated (man, mammals, birds).
- **Saprophytic (saprobial)** — feeding on dead and decaying matter by secreting enzymes (moulds, bacteria, mushrooms).
- **Parasitic** — feeding on the body of a living host, harming it (tapeworm, hookworm, plasmodium).
- **Symbiotic/mutualistic** — both partners benefit (the bacteria in the human gut producing vitamins; nitrogen-fixing bacteria in legume root nodules).

Feeding in simple organisms

- Amoeba feeds by engulfing food particles (phagocytosis); the food is enclosed in a food vacuole and digested by enzymes. Hydra paralyses prey with stinging cells (nematocysts).

CLASS EVALUATION

1. Define heterotrophic nutrition.
2. Distinguish between saprophytic and parasitic nutrition with examples.
3. How does an amoeba feed?
4. Give one example of mutualistic nutrition.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS1

SUBJECT: Biology

TOPIC: The Human Digestive System

Introduction

- Digestion is the breakdown of large, insoluble food molecules into small, soluble ones that can be absorbed. The human digestive system is a long tube (alimentary canal) with associated glands.

Structure in order

- **Mouth** — chewing and saliva (salivary amylase begins starch digestion).
- **Oesophagus** — peristalsis moves food to the stomach.
- **Stomach** — churns food; hydrochloric acid and pepsin digest proteins; the food becomes chyme.
- **Small intestine (duodenum and ileum)** — bile (from the liver) emulsifies fats; pancreatic juice completes digestion; digested food is absorbed through the villi.
- **Large intestine (colon)** — absorbs water and salts; **rectum and anus** — egestion of faeces.

Absorption and assimilation

- Villi and microvilli give a huge surface area; digested glucose and amino acids pass into the blood (to the liver via the hepatic portal vein); fatty acids enter the lacteals as chylomicrons. Assimilation is the use of absorbed food by body cells.

CLASS EVALUATION

1. Name the functions of the stomach and the small intestine.
2. What is the role of bile in digestion?
3. Why do villi increase absorption?
4. Distinguish between digestion, absorption and assimilation.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS1

SUBJECT: Biology

TOPIC: Enzymes & Their Role in Digestion

Introduction

- Enzymes are biological catalysts: protein molecules that speed up chemical reactions in living cells without being used up.

Properties of enzymes

- They are specific — each enzyme acts on one substrate (lipase on fats only).
- They work best at an optimum temperature (about 37 °C in humans) and optimum pH (pepsin in acid, trypsin in alkali).
- They are not used up and can be reused; extremes of heat (above 60–70 °C) denature them permanently.
- They work in small amounts and speed up both forward and backward reactions.

Enzymes in digestion

- **Salivary amylase (ptyalin)** — starch → maltose (mouth).
- **Pepsin** — proteins → peptones (stomach, acid medium).
- **Pancreatic amylase, trypsin and lipase** — starch, proteins and fats continue digestion (duodenum).
- **Maltase, sucrase, lactase** — sugars → glucose; **erepsin** — peptones → amino acids (ileum).

CLASS EVALUATION

1. Define an enzyme.
2. Explain what is meant by enzyme specificity.
3. What happens to pepsin in an alkaline medium and why?
4. State the substrate and product of salivary amylase.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS1

SUBJECT: Biology

TOPIC: Dentition & Feeding Habits

Introduction

- Dentition is the arrangement and type of teeth in the mouth. Teeth are adapted to the feeding habit of the animal: man is omnivorous (eats both plant and animal food).

Types of teeth and their functions

- **Incisors** — biting and cutting (8).
- **Canines** — tearing and holding food (4).
- **Premolars** — grinding and crushing (8).
- **Molars** — grinding and crushing (12, including the wisdom teeth).
- A child has 20 milk teeth; an adult has 32 permanent teeth (the dental formula for an adult is $I\ 2/2, C\ 1/1, Pm\ 2/2, M\ 3/2$).

Care of the teeth

- Brush twice daily with a fluoride toothpaste; floss between teeth; avoid excessive sweets and sugary drinks; visit the dentist regularly; eat foods that exercise the teeth (fruits, vegetables). Tooth decay (dental caries) results from acid formed by bacteria acting on sugar; gum disease (periodontitis) comes from plaque.

CLASS EVALUATION

1. State the function of each type of tooth.
2. How many permanent teeth does an adult have?
3. What causes dental caries?
4. Mention two ways of caring for the teeth.